



The Essentials of DeFi: Smart Contracts Not Banks

Decentralised Finance (DeFi) represents a fundamental shift in how financial services operate—built on **smart contracts** rather than traditional financial institutions, such as brokers

Unlike traditional finance with its middlemen, DeFi creates direct connections:

1

You

Control your assets directly

2

Smart Contract

Automated, code-based agreement

3

Other Person

Direct peer interaction

Centralised vs DeFi: The ETH to USDC Exchange

1

Centralised Exchange (CEX)

1. Send ETH from your wallet to Binance (commission)
2. Create sell order on exchange
3. Receive USDC after sale
4. Transfer USDC back to personal wallet (commission)

Your funds are held by the exchange throughout the process



2

Decentralized Exchange (DEX)

1. Connect wallet directly to Uniswap
2. Interact with swap smart contract (commission)
3. ETH is exchanged for USDC in one transaction

You maintain custody of your assets throughout



DeFi vs CeFi: The Core Differences

Feature	CeFi (Binance, Coinbase...)	DeFi (Aave, Uniswap...)
Custody	Exchange holds your funds	You hold your wallet keys
Transparency	Trust in company, Audits	On-chain, transparent smart contracts
Access	KYC/limits	No KYC
Risks	Exchange bankruptcy, hacks	Smart contract hacks

- Smart contracts can perform various financial functions - this variability enables different strategies for generating yields in the DeFi space. In next modules, you'll learn practical, real-world strategies to implement.

The **transparency** and **self-custody** aspects of DeFi represent its most revolutionary features, giving users unprecedented control over their financial activities.

Real-World DeFi Yield Opportunities

1

Lending Markets

Similar to traditional bank deposits, but with higher returns

- Deposit stablecoins like USDT/USDC into protocols like Aave or Compound
- Earn **5-15%** interest on your deposits
- Others borrow at **5-7%** rates, with the difference creating protocol reserves

2

Liquidity Provision (DEX/AMM)

Earn fees by supporting trading pairs on decentralized exchanges

- Deposit balanced pairs (e.g., 1,000 USDT + 1,000 USDC) into a liquidity pool
- Earn a percentage of each trade that uses your liquidity
- Yield often exceeds **10%** depending on trading volume

3

Liquidity Against Traders (Perpetuals)

Higher risk, higher reward option that benefits from trader liquidations

- Provide USDT as liquidity on perpetual DEXs like GMX or Hyperliquid
- Your balance grows when leveraged traders get liquidated
- Average yields can reach **~20%**

4

Staking

Support blockchain networks while earning native token rewards

- Stake tokens like SOL, ETH, or ATOM to secure their networks
- Earn network rewards, typically **~7% annually** in Sol as an example
- Lower risk but subject to token price volatility